

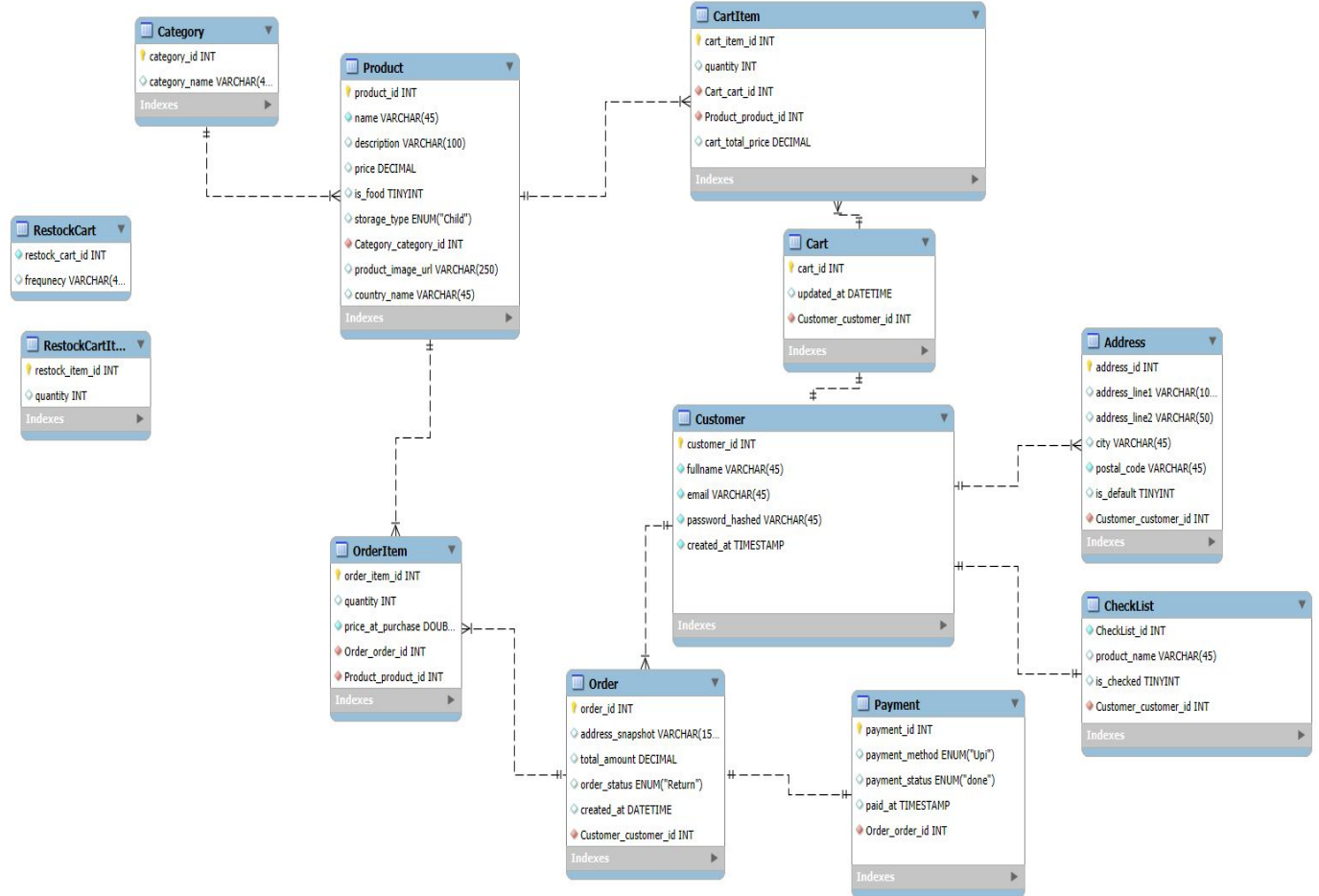


Internship Documents

ER Diagram

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ER-Diagram





- The ER model consists of 12 entities covering the complete e-commerce lifecycle.
- Core modules include User Management, Product Catalog, Cart System, Order Processing, Payments, and Inventory Restocking.
- One-to-Many relationships are used for:
 - Category–Product
 - Customer–Address
 - Cart–CartItem
 - Order–OrderItem

Category

- Stores product classification (e.g., Snacks, Beverages).
- Helps organize products for browsing and filtering.
- One category can contain multiple products.

Key Attributes

- `category_id` – Primary key
- `category_name` – Name of the category

Product

- Represents individual food items sold on the platform.
- Each product belongs to one category.
- Product availability can vary by country.

Key Attributes

- `product_id` – Primary key
- `name, description, price`
- `is_food, storage_type`
- `Product_image_url`
- `country_name`
- `category_id` – Foreign key

Customer

- Stores registered user details.
- Each customer belongs to one home country.
- A customer can place orders, maintain a cart, and store

Key Attributes

- `customer_id` – Primary key
- `fullname, email`
- `password_hashed`
- `created_at`

Address

- Stores delivery addresses for customers.
- A customer can have multiple addresses.
- Supports default address selection.

Key Attributes

- `address_id` – Primary key
- `address_line1`, `address_line2`
- `city`, `postal_code`
- `is_default`
- `customer_id` – Foreign key

Cart

- Represents the user's active shopping cart.
- Each customer has one active cart.
- Cart is temporary and changes frequently.

Key Attributes

- `cart_id` – Primary key
- `updated_at`
- `customer_id` – Foreign key

CartItem

- Stores products added to the cart.
- Acts as a link between Cart and Product.
- Quantity and total price are maintained here.

Key Attributes

- `cart_item_id` – Primary key
- `quantity`
- `cart_total_price`
- `cart_id, product_id` – Foreign keys

Order

- Represents a confirmed purchase.
- Created when the user checks out the cart.
- Stores final order amount and status.

Key Attributes

- `order_id` – Primary key
- `total_amount`
- `order_status`
- `created_at`
- `address`
- `customer_id` – Foreign key

Address

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Key Attributes

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- `city`, `postal_code`
- `is_default`
- `customer_id` – Foreign key

Cart

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Key Attributes

- `cart_id` – Primary key
- `updated_at`
- `customer_id` – Foreign key

OrderItem

- Stores individual products within an order.
- Captures price at the time of purchase.
- Prevents future price changes from affecting past orders.

Key Attributes

- `order_item_id` – Primary key
- `quantity`
- `price_at_purchase`
- `order_id, product_id` – Foreign keys

Payment

- Stores payment information for an order.
- Each order has one payment record.
- Supports multiple payment methods.

Key Attributes

- `payment_id` – Primary key
- `payment_method`
- `payment_status`
- `paid_at`
- `order_id` – Foreign key

RestockCart

- Used by users to add the frequent buying items.
- Represents a restocking session.

Key Attributes

- `restock_cart_id` – Primary key
- `frequency`

RestockCartItem

- Stores products and quantities to be restocked.
- Linked to RestockCart and Product.

Key Attributes

- `restock_item_id` – Primary key
- `quantity`
- `restock_cart_id` – Foreign key
- `product_id` – Foreign key

CheckList

- Used by users to make a checklist for shopping.
- Users add,remove, mark each item in the checklist.

Key Attributes

- `checklist_id` – Primary key
- `Product_name`
- `Is_checked`
- `Customer_id`– Foreign key